A counterexample to the three-shifted q-Motzkin divisibility conjecture

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Abstract

The size-five three-shifted Hankel determinant has value 2 at $q = -1$, while its conjectured divisor $1 + q^3$ vanishes. This disproves the divisibility part of the intended conjecture. The journal final remains unavailable for a corrections check; this proof therefore carries zero new campaign credit.

Task 1: the original conjecture and a counterexample

Write $M_n(q) = \widetilde{\mathcal{M}}_n^{\dagger}(q)$ for the polynomials defined on page 62 of [1] by

$$M_0(q) = 1, \qquad M_{n+1}(q) = M_n(q) + \sum_{r=0}^{n-1} q^{r+1} M_r(q) M_{n-r-1}(q).$$

Set

$$D_n(q) = \det[M_{i+j+3}(q)]_{i,j=0}^{n-1}, \qquad t_n(q) = \sum_{r=0}^{\lfloor n/3 \rfloor} q^{3r}.$$

Conjecture A.3, reproduced as MathDB 339077, intends $D_n = (-1)^{\lfloor n/3 \rfloor} q^{c_n} g_n(q)$, with $c_n \geq 0$, $g_n \in \mathbb{Z}_{\geq 0}[q]$, and in particular $t_n \mid g_n$ when $n \equiv 2 \pmod 3$. The printed display uses f_n where the ensuing text uses g_n. We resolve that notation in the intended way and disprove the substantive divisibility assertion.

Theorem 1. *We have $D_5(-1) = 2$. Consequently $1 + q^3$ does not divide $D_5(q)$, and no choice of monomial prefactor makes its normalized polynomial divisible by $t_5(q) = 1 + q^3$.*

Proof. Put $a_j = M_j(-1)$. Direct substitution in the defining recurrence gives

$$a_{n+1} = a_n + \sum_{r=0}^{n-1} (-1)^{r+1} a_r a_{n-r-1}.$$

Starting at $a_0 = 1$, its values through a_{11} are

$$(1, 1, 0, 0, 1, 1, -1, -1, 3, 3, -6, -6).$$

Thus $D_5(-1) = \det A$, where

$$A = \begin{pmatrix} 0 & 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 3 \\ 1 & -1 & -1 & 3 & 3 \\ -1 & -1 & 3 & 3 & -6 \\ -1 & 3 & 3 & -6 & -6 \end{pmatrix}.$$

Swap the first two rows, then eliminate the first two columns using their unit pivots. The remaining block is

$$B = \begin{pmatrix} 2 & 2 & -2 \\ 2 & 2 & -3 \\ -2 & -3 & 1 \end{pmatrix}, \qquad \det A = -\det B = 2.$$

Indeed, subtracting the first row of B from its second and adding it to its third leaves lower rows $(0, 0, -1)$, $(0, -1, -1)$; hence $\det B = -2$. If $t_5 \mid g_5$ and $D_5 = -q^{c_5} g_5$, then evaluation at -1 would give $D_5(-1) = 0$, a contradiction. $\square$

Scope and exact checks

Evaluation at a negative value of q is legitimate: the claim is polynomial divisibility, so a proposed identity must hold under every specialization. The argument does not assume that q is a positive probability parameter and does not exploit the f_n/g_n notation. A single admissible size $5 \equiv 2 \pmod 3$ disproves the universal claim.

As a supplemental certificate, exact polynomial elimination gives

$$D_5(q) = -q^{62}\big(1 + 2q + 2q^3 + 2q^4 + q^6 + 2q^7\big).$$

Its remainder on division by $1+q^3$ is the constant 2. The elementary proof above needs only the twelve displayed integers and the row operations, not this larger polynomial computation.

The verifier checks the integer recurrence, the determinant by all 120 signed permutation products and by the displayed elimination, the full polynomial determinant and its remainder, and twelve moment polynomials by a separate weighted-walk computation. These are exact checks of a finite counterexample, not numerical evidence for an unbounded assertion.

All 68 pages of the current primary preprint, including the references, were read. Its page 62 was also inspected visually. The 2019 journal final [2] was identified but could not be obtained in full; possible corrections and the complete subsequent literature remain an eligibility limitation. The earlier repository package for MathDB 339076 evaluates shift two and proposes shift three as a follow-up; it does not supply this counterexample. No independent peer review or proof-assistant certification is claimed.

Task 2: determine the correct divisors

Classify the sizes $n \equiv 2 \pmod 3$ for which $t_n \mid D_n$. Compute the exact greatest common divisor with t_n in the remaining cases, using roots of unity to isolate the obstructing factors.

Task 3: evaluate all three-shift determinants

Find a uniform formula for D_n, including its order at $q = 0$. Separate the surviving coefficient-sign and square-factor assertions from the divisibility assertion disproved here.

Task 4: root-of-unity specializations

Find recurrences for $M_n(\zeta)$ and $D_n(\zeta)$ when ζ has finite multiplicative order. Determine whether the size-five obstruction extends to an infinite family of sizes.

Task 5: higher shifts

Investigate coefficient signs and cyclotomic factors of $\det[M_{i+j+s}(q)]$ as the shift s varies. In particular, the separate all-even-shifts positivity conjecture, MathDB 339078, is not settled by this counterexample. Tasks 2–5 are proposed directions and carry no additional credit.

References

[1] S.-j. Oh and T. Scrimshaw, *Identities from representation theory*, arXiv:1805.00113v1 (2018), Appendix A, pp. 61–63. `https://arxiv.org/abs/1805.00113v1`.

[2] S.-j. Oh and T. Scrimshaw, *Identities from representation theory*, Discrete Mathematics **342**(9) (2019), 2493–2541. `https://doi.org/10.1016/j.disc.2019.05.020`. Bibliographic identification only; final full text not read.